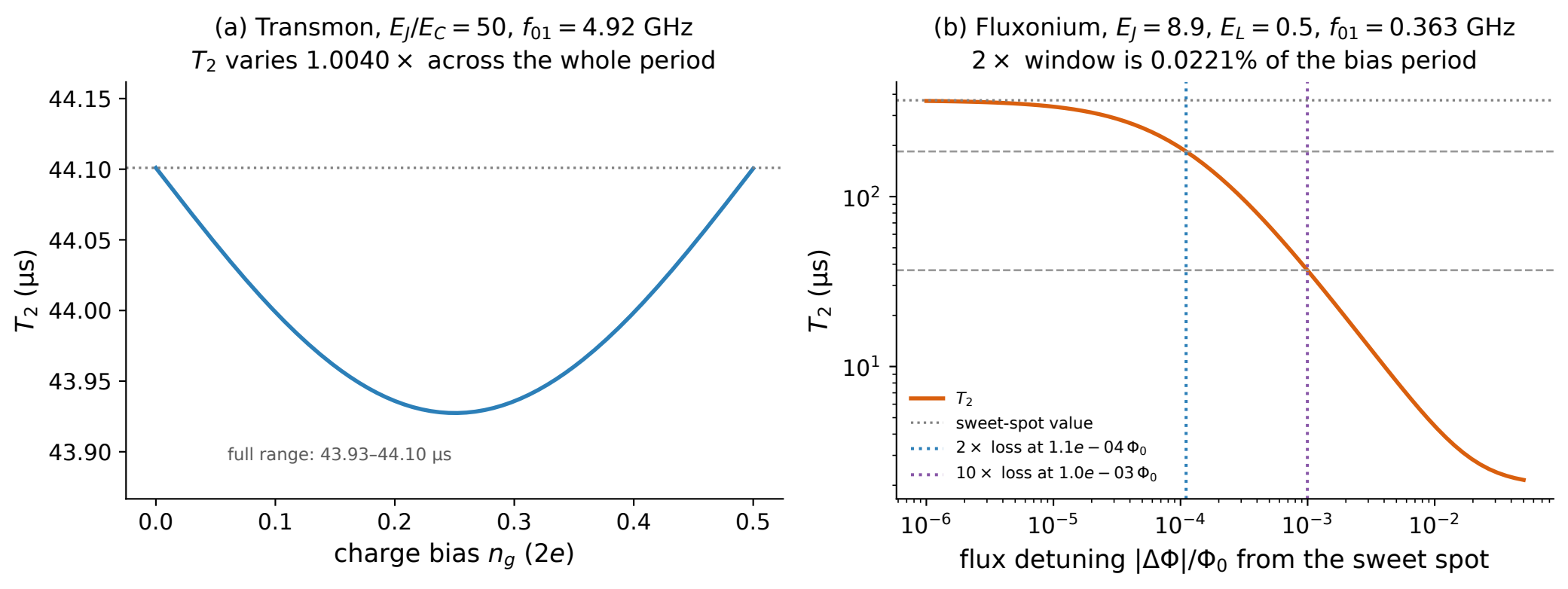

(a) Transmon, $E_J/E_C = 50$, $f_{01} = 4.92$ GHz
T_2 varies 1.0040 × across the whole period
44.15
44.10
44.05
44.00
43.95
43.90
T_2 (μs)
full range: 43.93–44.10 μs
0.0
0.1
0.2
0.3
0.4
0.5
charge bias n_g (2e)
(b) Fluxonium, $E_J = 8.9$, $E_L = 0.5$, $f_{01} = 0.363$ GHz
2 × window is 0.0221% of the bias period
10^2
10^1
T_2 (μs)
T_2
sweet-spot value
2 × loss at $1.1e-04\,\Phi_0$
10 × loss at $1.0e-03\,\Phi_0$
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
flux detuning $|\Delta\Phi|/\Phi_0$ from the sweet spot